

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 410 — Past Questions & Materials
by Renance

Course: CHM 410
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — CHM 410 Analytical Chemistry

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN DELTA STATE

DEPARTMENT OF CHEMISTRY

FIRST SEMESTER EXAMINATION 2024/2025 SESSION

COURSE CODE: CHM410; COURSE TITLE: ANALYTICAL CHEMISTRY

TIME ALLOWED: 2 HOURS

INSTRUCTIONS: ANSWER FOUR (4) QUESTIONS ONLY

- 1a. State the principles of chromatography. (3marks)
 - b. Substances A and B have retention times of 16.40 and 17.63 min, respectively, on a 30.0cm column. An unretained specie passes through the column in 1.30min. The peak widths (at base) for A and B are 1.11 and 1.21min, respectively. Calculate (i) the column resolution (ii) the average number of plates in the column, (iii) the plate height. (9marks)
 - c. Define (i) elution (ii) retention time and (iii) selectivity factor (6marks)
- 2a. What is a chromatogram? (4marks)
- b. How do the gas-liquid and gas-solid chromatography differs? (4marks)
 - c. Why is gas-solid chromatography not used nearly as extensively as gas-liquid chromatography? (4marks)
 - d. A component in a mixture was separated by chromatographic procedure. The retention time for the component is 8.5min and the base width of the peak 1.25min. Calculate the efficiency of the separation of this peak in terms of its plate number, N. (5marks)
- 3a. The refractive index (u) of water is found to have the values 1.29, 1.33, 1.34, 1.35, 1.32, 1.36, 1.30 and 1.33. Calculate the mean value, absolute error, the relative error and percentage error. (7 marks)
- b. Suppose you measure the following weights of some samples
- | Samples | Measurement (g) |
|---------|-----------------|
| 1 | 1000 |
| 2 | 1010 |
| 3 | 1015 |
| 4 | 850 |
- (i) Determine at $Q_{0.90}$ if the fourth measurement should be accepted or treated as an outlier (2.5 marks)

- (ii) Calculate the mean and determine the standard deviation (4 marks)
- (iii) What are the confidence limit and the confidence interval? (2 marks)
- c. Briefly differentiate between accuracy and precision (2 marks).
- 4a. State Beer- Lambert law (4 marks)
- b. The absorption coefficient of a glycogen-iodine complex is 0.20 at wavelength of 450 nm. What is the concentration when the transmission is 40% in a cuvette of 2 cm? (4marks)
- c. There is a substance in a solution (4 g/L). The length of cuvette is 2 cm and only 50% of the certain beam is transmitted. What is the extinction coefficient? (4marks)
- d. A constant current of 0.90 A is used to deposit copper at the cathode and oxygen at the anode of an electrolytic cell. Calculate the number of grams of each product formed in 17.2 min, assuming no other redox reaction occurs (5marks) $M_{Cu} = 63.5$, $O = 16$, $F = 96500$
- 5a. What is radiochemical method of analysis (2.5marks)
- b. A sample of hair from an Egyptian mummy gives off radiation from Carbon-14 at a rate of 58.2% of a present-day sample. How old is the mummy if the half-life of carbon-14 is 5730 years (5marks)
- c. Explain Isotope dilution analysis (6 marks)
- d. Give four applications of controlled potential coulometry (4marks)
- 6a. State the principle of potentiometry (5marks)
- b. Calculate the formation constant K, for $Ag(CN)_2^-$: $Ag^+ + 2CN^- \rightleftharpoons Ag(CN)_2^-$, If the cell $SCE || Ag(CN)_2 (7.50 \times 10^{-3} M), CN^- (0.0250 M) | Ag$ develops a potential of -0.625 V. ($E^0 = +0.799$ V, $E_{Ag^+} = -0.244$) (4.5marks)
- c. The half-life of 1.00g of radium is 1500 years. Find (i) the decay constant in units of per seconds (ii) the number of radium atoms (iii) the activity (iv) specific activity (Ra=226, Avogadro's constant= 6.02×10^{23} particles) (8marks)

$$n \approx \text{moles} \times \text{Avogadro's number} \quad \int \frac{1}{2} = \frac{0.693}{1}$$

Paper 2 of 2 — CHM 410 ANALYTICAL CHEMISTRY 2024 2025

Level 400 • First Semester • PastQuestion

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$$n \text{ or moles} \times \text{Avogadro's number} \cdot t_{1/2} = \frac{0.693}{\lambda}$$

QUESTION 7

- (a) (i) Given the point $A(4, 1)$ and the vector $V = 7i - 3j$. Find $B =$
? and draw the points on a plane
- (ii) Find the value of θ if $U = \langle 2, -1, -8 \rangle$ and $V = \langle 3, 5, 1 \rangle$
- (iii) Given the function $f(x, y) = xy^2 + 3x^3$ in the direction given by $\theta = \frac{\pi}{6}$.
Evaluate the gradient vector at the point $(2, -3)$
- (b) Evaluate the directional derivative at the same point.